operationalism, 174–175, 177, 184, 185, 187
Operation Alsos, 75–76, 77, 81
Oppenheimer, Robert, 73, 85, 90, 93, 94, 114, 133, 247

Pan, Jianwei, 267
paradigms, 182–183
paradox, 68, 98, 118, 127
  time travel paradoxes, 25, 152. *See also* relativity, speed of light
particle accelerators, 111, 144, 228, 286
particles/waves. *See* wave-particle duality
Patterns of Discovery (Hanson), 183
Pauli, Wolfgang, 13, 23, 34, 35(photo), 40, 43, 44(photo), 45–46, 48, 49, 57, 93, 96, 103, 109, 110, 164, 173, 174, 283
  and Bohm's ideas, 107
  philosophy of science of, 172
Pearle, Philip, 237
Peat, David, 91
Peierls, Rudolf, 33–34, 74, 144
Penrose, Roger, 286
Pentagon, 120, 250
  Weapons Systems Evaluation Group (WSEG), 131, 135
Penzias, Arno, 248
periodic table of elements, 2, 13, 142, 183
Petersen, Aage, 121, 122, 129, 131, 132–133, 164
Philippidis, Chris, 227
Phillips, Melba, 92
philosophy, 82, 167, 170, 179, 189, 211, 214, 261
  analytic vs. Continental, 273–274, 327–328, 340
  and physicists, 272–273, 288
  See also science: philosophy of science
photoelectric effect, 27
photons, 27–28, 101–102, 176, 195, 267
  entangled, 149, 150, 151, 156–157, 160, 205, 215–216, 233, 268
  delayed-choice thought experiment, 289–294
  See also double-slit experiment
photosynthesis, 27